Part 1

main.cpp

Clear Run

```cpp

#include <iostream>
using namespace std;

template <typename T>
class Stack {
private:
    T* arr;
    int capacity;
    int topIndex;

public:
    Stack() {
        capacity = 2;
        arr = new T[capacity];
        topIndex = -1;
    }

    ~Stack() {
        delete[] arr;
```

Output

```
Empty stack: 1
Stack size: 2
Top element: 20
Stack size after pushing 30: 3
Popped: 30
Popped: 20
Popped: 10
Stack empty: 1
```

main.cpp

Clear Run

```cpp
    // Test pop
    cout << "\nPopping elements:" << endl;

    while (myStack.pop(value)) {
        cout << "Popped: " << value << endl;
    }

    // Test empty stack again
    cout << "\nFinal empty: "
         << myStack.isEmpty() << endl;

    cout << "Final peek: "
         << myStack.peek(value) << endl;

    cout << "Final size: "
         << myStack.size() << endl;

    return 0;
}

```

Output

```
===== STACK SAFETY TEST =====
Initial empty: true
Pop empty stack: false
Peek empty stack: false
Size: 0

After pushing 10 and 20:
Size: 2
Peek: 20

After pushing 30:
Size: 3

Popping elements:
Popped: 30
Popped: 20
Popped: 10

Final empty: true
Final peek: false
```

main.cpp

Clear Run

```cpp
    // Test pop
    cout << "\nPopping elements:" << endl;

    while (myStack.pop(value)) {
        cout << "Popped: " << value << endl;
    }

    // Test empty stack again
    cout << "\nFinal empty: "
         << myStack.isEmpty() << endl;

    cout << "Final peek: "
         << myStack.peek(value) << endl;

    cout << "Final size: "
         << myStack.size() << endl;

    return 0;
}

```

Output

```
Pop empty stack: false
Peek empty stack: false
Size: 0

After pushing 10 and 20:
Size: 2
Peek: 20

After pushing 30:
Size: 3

Popping elements:
Popped: 30
Popped: 20
Popped: 10

Final empty: true
Final peek: false
Final size: 0
```

Part 2

main.cpp

```
132
133     expression = "{[2 + (3 * 4)]}";
134     cout << "\nInput: " << expression << endl;
135     cout << "Result: " << checkBrackets(expression) << endl;
136
137     expression = "([)]";
138     cout << "\nInput: " << expression << endl;
139     cout << "Result: " << checkBrackets(expression) << endl;
140
141     expression = "(2 + 3";
142     cout << "\nInput: " << expression << endl;
143     cout << "Result: " << checkBrackets(expression) << endl;
144
145     expression = "2 + 3)";
146     cout << "\nInput: " << expression << endl;
147     cout << "Result: " << checkBrackets(expression) << endl;
148
149     return 0;
150 }
151
```

Output

```
===== CC105 WEEK 8 LABORATORY =====

===== PART 2: BRACKET CHECKER =====

Input: {[2 + (3 * 4)]}
Result: Balanced

Input: ([)]
Result: Mismatched brackets

Input: (2 + 3
Result: Unclosed opening bracket

Input: 2 + 3)
Result: Unmatched closing bracket
```

Part 3

main.cpp

```
317
318         cout << "Bracket Status: " << bracketStatus << endl;
319
320         if (bracketStatus != "Balanced") {
321             cout << "Error: " << bracketStatus << endl;
322             continue;
323         }
324
325         if (infixToPostfix(expression, postfix, error)) {
326             cout << "Postfix: " << displayPostfix(postfix) << endl;
327             cout << "Status: Valid conversion" << endl;
328         }
329         else {
330             cout << "Error: " << error << endl;
331         }
332     }
333
334     return 0;
335 }
336
```

Output

```
===== CC105 WEEK 8 LABORATORY =====
===== PART 3: INFIX TO POSTFIX =====

Input: (12 + 3) * 2
Bracket Status: Balanced
Postfix: 12 3 + 2 *
Status: Valid conversion

Input: 8 + 2 * 5
Bracket Status: Balanced
Postfix: 8 2 5 * +
Status: Valid conversion

Input: 20 / 5 / 2
Bracket Status: Balanced
Postfix: 20 5 / 2 /
Status: Valid conversion

Input: 10 - 3 - 2
Bracket Status: Balanced
```

main.cpp

```
317
318         cout << "Bracket Status: " << bracketStatus << endl;
319
320         if (bracketStatus != "Balanced") {
321             cout << "Error: " << bracketStatus << endl;
322             continue;
323         }
324
325         if (infixToPostfix(expression, postfix, error)) {
326             cout << "Postfix: " << displayPostfix(postfix) << endl;
327             cout << "Status: Valid conversion" << endl;
328         }
329         else {
330             cout << "Error: " << error << endl;
331         }
332     }
333
334     return 0;
335 }
336
```

Output

```
Postfix: 10 3 - 2 -
Status: Valid conversion

Input: {[2 + 3] * (4 + 1)}
Bracket Status: Balanced
Postfix: 2 3 + 4 1 + *
Status: Valid conversion

Input: 7 / 2
Bracket Status: Balanced
Postfix: 7 2 /
Status: Valid conversion

Input: 2 + * 3
Bracket Status: Balanced
ERROR!
Error: Invalid operator sequence or unary sign

Input: 2 3 + 4
Bracket Status: Balanced
```

main.cpp Clear Run ▶ Output

```
317
318         cout << "Bracket Status: " << bracketStatus << endl;
319
320         if (bracketStatus != "Balanced") {
321             cout << "Error: " << bracketStatus << endl;
322             continue;
323         }
324
325         if (infixToPostfix(expression, postfix, error)) {
326             cout << "Postfix: " << displayPostfix(postfix) << endl;
327             cout << "Status: Valid conversion" << endl;
328         }
329         else {
330             cout << "Error: " << error << endl;
331         }
332     }
333
334     return 0;
335 }
336
```

```
Input: 2(3 + 4)
Bracket Status: Balanced
Error: Implicit multiplication rejected

Input: ()
Bracket Status: Balanced
Error: Empty group or missing operand

Input: -5 + 2
Bracket Status: Balanced
Error: Invalid operator sequence or unary sign

Input: 2.5 + 1
Bracket Status: Balanced
Error: Decimal numbers are not supported

Input: 2 + a
Bracket Status: Balanced
Error: Unsupported character
```

Part 4

```
369 int main() {
370     cout << "===== CC105 WEEK 8 LABORATORY =====" << endl;
371     cout << "===== PART 4: POSTFIX EVALUATION =====" << endl;
372
373     vector<string> tests = {
374         "(12 + 3) * 2",
375         "8 + 2 * 5",
376         "20 / 5 / 2",
377         "10 - 3 - 2",
378         "{[2 + 3] * (4 + 1)}",
379         "7 / 2",
380         "0",
381         "5 / (3 - 3)",
382         "2 + * 3",
383         "2 3 + 4",
384         "2(3 + 4)",
385         "()",
386         "-5 + 2",
387         "2.5 + 1",
388         "2 + a"
389     };
```

```
===== CC105 WEEK 8 LABORATORY =====
===== PART 4: POSTFIX EVALUATION =====

Input: (12 + 3) * 2
ERROR!
Bracket Status: Balanced
Postfix: 12 3 + 2 *
Result: 30
Status: Valid expression

Input: 8 + 2 * 5
Bracket Status: Balanced
Postfix: 8 2 5 * +
Result: 18
Status: Valid expression

Input: 20 / 5 / 2
Bracket Status: Balanced
Postfix: 20 5 / 2 /
Result: 2
```

main.cpp Clear Run ▶ Output

```
369 int main() {
370     cout << "===== CC105 WEEK 8 LABORATORY =====" << endl;
371     cout << "===== PART 4: POSTFIX EVALUATION =====" << endl;
372
373     vector<string> tests = {
374         "(12 + 3) * 2",
375         "8 + 2 * 5",
376         "20 / 5 / 2",
377         "10 - 3 - 2",
378         "{[2 + 3] * (4 + 1)}",
379         "7 / 2",
380         "0",
381         "5 / (3 - 3)",
382         "2 + * 3",
383         "2 3 + 4",
384         "2(3 + 4)",
385         "()",
386         "-5 + 2",
387         "2.5 + 1",
388         "2 + a"
```

```
Status: Valid expression

Input: 10 - 3 - 2
Bracket Status: Balanced
Postfix: 10 3 - 2 -
Result: 5
Status: Valid expression

Input: {[2 + 3] * (4 + 1)}
Bracket Status: Balanced
Postfix: 2 3 + 4 1 + *
Result: 25
Status: Valid expression

Input: 7 / 2
Bracket Status: Balanced
Postfix: 7 2 /
Result: 3.5
Status: Valid expression
```

main.cpp

Clear Run

```cpp
int main() {
    cout << "===== CC105 WEEK 8 LABORATORY =====" << endl;
    cout << "===== PART 4: POSTFIX EVALUATION =====" << endl;

    vector<string> tests = {
        "(12 + 3) * 2",
        "8 + 2 * 5",
        "20 / 5 / 2",
        "10 - 3 - 2",
        "{[2 + 3] * (4 + 1)}",
        "7 / 2",
        "0",
        "5 / (3 - 3)",
        "2 + * 3",
        "2 3 + 4",
        "2(3 + 4)",
        "()",
        "-5 + 2",
        "2.5 + 1",
        "2 + a"
    };
```

Output

```
Input: 0
Bracket Status: Balanced
Postfix: 0
Result: 0
Status: Valid expression

Input: 5 / (3 - 3)
Bracket Status: Balanced
Postfix: 5 3 3 - /
Error: Division by zero

Input: 2 + * 3
Bracket Status: Balanced
Error: Invalid operator sequence or unary sign

Input: 2 3 + 4
Bracket Status: Balanced
Error: Missing operator

Input: 2(3 + 4)
```

main.cpp

Clear Run

```cpp
int main() {
    cout << "===== CC105 WEEK 8 LABORATORY =====" << endl;
    cout << "===== PART 4: POSTFIX EVALUATION =====" << endl;

    vector<string> tests = {
        "(12 + 3) * 2",
        "8 + 2 * 5",
        "20 / 5 / 2",
        "10 - 3 - 2",
        "{[2 + 3] * (4 + 1)}",
        "7 / 2",
        "0",
        "5 / (3 - 3)",
        "2 + * 3",
        "2 3 + 4",
        "2(3 + 4)",
        "()",
        "-5 + 2",
        "2.5 + 1",
        "2 + a"
    };
```

Output

```
Input: 2(3 + 4)
Bracket Status: Balanced
Error: Implicit multiplication rejected

Input: ()
Bracket Status: Balanced
Error: Empty group or missing operand

Input: -5 + 2
Bracket Status: Balanced
Error: Invalid operator sequence or unary sign

Input: 2.5 + 1
Bracket Status: Balanced
Error: Decimal numbers are not supported

Input: 2 + a
Bracket Status: Balanced
Error: Unsupported character
```

Part 5

main.cpp

Clear Run

```cpp
        }
        else if (choice == 2) {
            expressionMenu();
        }
        else if (choice == 3) {
            stackDemo();
        }
        else if (choice == 0) {
            cout << "\nThank you for using the program!" << endl;
            break;
        }
        else {
            cout << "\nInvalid choice. Please select 0, 1, 2, or
3." << endl;
        }
    }

    return 0;
}
```

Output

```
==================================
       CC105 WEEK 8 LABORATORY
==================================
1. Check Brackets
2. Convert and Evaluate Expression
3. Stack Demonstration
0. Exit
==================================
Enter your choice: 1

Enter an expression: 1
Bracket Status: Balanced

==================================
       CC105 WEEK 8 LABORATORY
==================================
1. Check Brackets
2. Convert and Evaluate Expression
3. Stack Demonstration
0. Exit
```

main.cpp

Clear Run

```
            }
        else if (choice == 2) {
            expressionMenu();
        }
        else if (choice == 3) {
            stackDemo();
        }
        else if (choice == 0) {
            cout << "\nThank you for using the program!" << endl;
            break;
        }
        else {
            cout << "\nInvalid choice. Please select 0, 1, 2, or
3." << endl;
        }
    }

    return 0;
}
```

Output

```
Enter your choice: 2

Enter an arithmetic expression: 2

Input: 2
Bracket Status: Balanced
Postfix: 2
Result: 2
Status: Valid expression

====================================
       CC105 WEEK 8 LABORATORY
====================================
1. Check Brackets
2. Convert and Evaluate Expression
3. Stack Demonstration
0. Exit
====================================
Enter your choice: 3
```

main.cpp

Clear Run

```
            }
        else if (choice == 2) {
            expressionMenu();
        }
        else if (choice == 3) {
            stackDemo();
        }
        else if (choice == 0) {
            cout << "\nThank you for using the program!" << endl;
            break;
        }
        else {
            cout << "\nInvalid choice. Please select 0, 1, 2, or
3." << endl;
        }
    }

    return 0;
}
```

Output

```
===== STACK DEMONSTRATION =====
Initial empty: true
Pop empty stack: false

Pushing 10, 20, and 30...
Stack size: 3
Top element: 30

Popping elements:
Popped: 30
Popped: 20
Popped: 10
Final empty: true
Final size: 0

====================================
       CC105 WEEK 8 LABORATORY
====================================
1. Check Brackets
```

main.cpp

Clear Run

```
            }
        else if (choice == 2) {
            expressionMenu();
        }
        else if (choice == 3) {
            stackDemo();
        }
        else if (choice == 0) {
            cout << "\nThank you for using the program!" << endl;
            break;
        }
        else {
            cout << "\nInvalid choice. Please select 0, 1, 2, or
3." << endl;
        }
    }

    return 0;
}
```

Output

```
Popping elements:
Popped: 30
Popped: 20
Popped: 10
Final empty: true
Final size: 0

====================================
       CC105 WEEK 8 LABORATORY
====================================
1. Check Brackets
2. Convert and Evaluate Expression
3. Stack Demonstration
0. Exit
====================================
Enter your choice: 0

Thank you for using the program!
```

main.cpp

Clear | Run ▶

```
500         }
501         else if (choice == 2) {
502             expressionMenu();
503         }
504         else if (choice == 3) {
505             stackDemo();
506         }
507         else if (choice == 0) {
508             cout << "\nThank you for using the program!" << endl;
509             break;
510         }
511         else {
512             cout << "\nInvalid choice. Please select 0, 1, 2, or
    3." << endl;
513         }
514     }
515
516     return 0;
517 }
518
```

Output

```
==================================
       CC105 WEEK 8 LABORATORY
==================================
1. Check Brackets
2. Convert and Evaluate Expression
3. Stack Demonstration
0. Exit
==================================
Enter your choice: 2

Enter an arithmetic expression: ((2 + 3) * (4 + 5))

Input: ((2 + 3) * (4 + 5))
Bracket Status: Balanced
Postfix: 2 3 + 4 5 + *
Result: 45
Status: Valid expression

==================================
```

main.cpp

Clear | Run ▶

```
500         }
501         else if (choice == 2) {
502             expressionMenu();
503         }
504         else if (choice == 3) {
505             stackDemo();
506         }
507         else if (choice == 0) {
508             cout << "\nThank you for using the program!" << endl;
509             break;
510         }
511         else {
512             cout << "\nInvalid choice. Please select 0, 1, 2, or
    3." << endl;
513         }
514     }
515
516     return 0;
517 }
518
```

Output

```
==================================
       CC105 WEEK 8 LABORATORY
==================================
1. Check Brackets
2. Convert and Evaluate Expression
3. Stack Demonstration
0. Exit
==================================
Enter your choice: 2

Enter an arithmetic expression: 1 + 2 + 3 + 4 + 5 + 6 + 7 + 8 + 9 + 10

Input: 1 + 2 + 3 + 4 + 5 + 6 + 7 + 8 + 9 + 10
Bracket Status: Balanced
Postfix: 1 2 + 3 + 4 + 5 + 6 + 7 + 8 + 9 + 10 +
Result: 55
Status: Valid expression

==================================
```

main.cpp

Clear | Run ▶

```
500         }
501         else if (choice == 2) {
502             expressionMenu();
503         }
504         else if (choice == 3) {
505             stackDemo();
506         }
507         else if (choice == 0) {
508             cout << "\nThank you for using the program!" << endl;
509             break;
510         }
511         else {
512             cout << "\nInvalid choice. Please select 0, 1, 2, or
    3." << endl;
513         }
514     }
515
516     return 0;
517 }
518
```

Output

```
Bracket Status: Balanced
Postfix: 1 2 + 3 + 4 + 5 + 6 + 7 + 8 + 9 + 10 +
Result: 55
Status: Valid expression

==================================
       CC105 WEEK 8 LABORATORY
==================================
1. Check Brackets
2. Convert and Evaluate Expression
3. Stack Demonstration
0. Exit
==================================
Enter your choice: 2

Enter an arithmetic expression: 5 + * 8
ERROR!

Input: 5 + * 8
Bracket Status: Balanced
```